Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	Hunter, Gary Ph.D.		
Project Title:	Microsystems-Based Hydrazine Detection System for International Space Station and EVA Applications		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-01
Start Date:	07/15/2002	**End Date:**	07/15/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hong , Todd (Munez Engineering Inc./NASA Johnson Space Center) Young, Rebecca (NASA Kennedy Space Center) Makel, Darby (Makel Engineering, Inc.) Liu, C. (Case Western Reserve University)		
Grant/Contract No.:			
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Task Description:	The detection of hydrazine at low concentrations (down to 10 ppb) is important for International Space Station Environmental Monitoring and EVA (Extra Vehicular Activity) applications. Low concentration of hydrazine can cause serious health problems for astronauts who can be exposed to hydrazine while on EVA mission outside of the International Space Station. Early detection of hydrazine is required both during an EVA, in the airlock, and as the airlock is opened into the Station. These applications require a potential hydrazine sensor to provide real-time response in vacuum, during pressurization, and in the presence of interfering gases. Present technology is not adequate for use in this Space Station application. Deficiencies include: 1. Inability to operate in the ISS environments; 2. Existing systems are too large for use where needed e.g. on the space suit; 3. Susceptibility to false alarm due, in part, to the presence of interfering gases; 4. Sensor has long response time or is hard to read; 5. Monitoring device does not interface readily to electronics to provide early warning signals. This task seeks to develop a microfabricated hydrazine sensor system to detect hydrazine for Space Station Environmental Monitoring and EVA applications. This sensor system will be developed using Schottky diode technology that has been shown to be effective in similar operational environments esp. leak detection on the Space Shuttle. A Schottky diode is a semiconductor-metal contact with possibly a thin interface layer in-between the metal and the semiconductor. Major challenges associated with this project are the very low concentrations measurements necessary and the need to measure in the presence of interfering gases. However, Schottky diode gas sensors have exponential responses, do not require oxygen for their operation, and are thus very well suited to this application. Various approaches will be investigated this project including the use of silicon carbide (SiC) as the semiconductor as well as various metal contacts and interface layers. Even if a sensor has short-term response the hydrazine, that does not mean that the sensor has long-term sensitivity and stability in a given environment. Thus, along with new approaches to hydrazine detection, long-term stability studies are necessary. Since interfering gas are present which could make determination of the exact hydrazine concentration difficult for a single sensor, the hydrazine sensor will be placed in an array. Leveraging other sensor and electronics development will be done as appropriate to produce a hydrazine sensor system to meet a significant Space Station technology need which presently cannot be addressed. This task brings together a highly qualified team which has a high degree of knowledge of the application combined with considerable experience in chemical sensor development, characterization, and application. The result of this project will not be flight hardware but prototype technology which performs in environments where other sensor technology is inadequate. This technology would be designed with flight consideration in mind and could later be qualified for flight use but that is not part of this present project. The purpose of this work is to improve astronaut health and safety by providing low concentration hydrazine detection with the following properties as previously determined by JSC personnel: 25 ppb minimum measurement of Hydrazine 5 minute response time 5 psi pressure operation Lack of corresponding response to ammonia. Improved monitoring of toxic materials such as hydrazine is important for present applicaitons such as International Space Station EVAs. Contaminant monitoring will continue to be mandatory in future exploration systems where contamination of closed cycled vehicle systems could have significant effects on astronaut health in long-term missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The detection of low concentrations of toxic chemicals has a broad range of earth related applications. While this technology is being developed specifically for hydrazine detection, its base technology has application to a range of other toxic chemicals as well.
Task Progress:	The task has continued to show the feasibility of the approach of using a Schottky diode gas sensor for this application. Testing has been ongoing to verify system capability to meet suggested performance requirements. This includes operation of a prototype sensor with sensitivity down to 11 ppb MMH with a strong signal has been demonstrated with response times appropriate for the application. Response for 90% full signal within 5 minutes has been achieved. More recent work has concentrated on determining if there cross-sensitivity in the present sensor response between ammonia and MMH. Preliminary testing has shown that the sensor has minimal response (basically slight drift) to over 100 ppm of ammonia. The magnitude of that minimal is very small compared to the magnitude of response for 25 ppb levels of MMH. Given that levels for ammonia (according to SMAC requirements) that present health concern are less 100 ppm ammonia, we have been advised that these results suggest a lack of cross-sensitivity to ammonia for expected environment of this application. However, the sensor response to MMH is changed slightly by ammonia exposure. In fact, it becomes slightly more sensitive to MMH after ammonia exposure if the tests are done in near succession. It appears that ammonia "cleans" the surface to allow more sensitivity to MMH. A significant number of ammonia followed by MMH tests followed by partial pressure air exposure has been done to verify the repeatability of the sensor response over multiple ammonia/MMH exposures. The results suggest some effect is evident but new samples have been prepared for further testing. Nonetheless, given the small magnitude of the effect, it appears that a companion ammonia sensor (for which we have candidates) combined with the hydrazine sensor may be a viable method to address this issue even in the worse case scenario. The level at which this combination sensor approach will explored will depend on the results of the future testing of modified versions of the Schottky diode sensor. Software augmentation of the data signal has also been on-going. Using an algorithm based on the slope of the signal, the response time can be greatly reduced i.e. to times of near a minute. Various parameters in the software can be set in to maximize the initial software alerts to a detection of hydrazine. However, if these parameters are not properly set, the software calculation of the hydrazine concentration may initially overestimate the actual concentration. Thus, the algorithm must be designed with a balance between the amount of hydrazine concentration over estimate tolerable and the response time required. Further, sensor exposure to ammonia in the absence of MMH produces a very small signal. On the other hand, adding ammonia to a given MMH level seems to produce a cumulative effect and seems to speed up the initial response. Applying the calibration and speed up algorithm, it may be possible to differentiate between pure MMH and MMH/ammonia mix. For example, if the initial response is really, there is likely a mix of MMH/ammonia, or a very high MMH content (e.g. <50 ppb). If the steady state response (a few minutes) does not show a large concentration, a MMH/ammonia mix presence is likely. These software approaches, while at their beginning stages, are meant to improve the overall response of the sensor system.

	Future work includes determination of longer-term hydrazine sensor stability/repeatability and integration of software in sensor system hardware to improve species identification and detection time.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Other Journals or Periodicals	Liu CC, Hesketh PJ, Hunter GW. "Chemical microsensors." Interface Magazine, 2004 Summer. p 22. , Jun-2004
Articles in Peer-reviewed Journals	Hunter,G.W.; Neudeck, P.G.; Xu, J.; Lukco, D.; Trunek, A.; Artale, M.; Lampard, P.; Androjna, D.; Makel, D.; Liu, C.; "Development of SiC-based Gas Sensors for Aerospace Applications" Materials Research Society Symposium Proceedings, vol. 815, Silicon Carbide 2004 - Materials, Processing, and Devices, M. Dudley, P. Gouma, T. Kimoto, P. G. Neudeck, and S. E. Saddow Eds. Warrendale, PA: Materials Research Society, 2004, pp. 287-297. Also, invited paper in Spring 2004 MRS meeting, May 2004 , Aug-2004
Presentation	Hunter, G.W.; "MICROFABRICATED CHEMICAL SENSORS FOR" Presentation to class in “Molecular Engineering of Microscale Devices“ which is part of the National Science Foundation program for “Integrated graduate education, research and traineeship“ (IGERT). Apr-2004
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